

OMNIWATCH OS

Intelligence, Surveillance & Reconnaissance
Command & Control Platform

Comprehensive Product Documentation

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1. Executive Summary

OMNIWATCH OS is a next-generation Intelligence, Surveillance, and Reconnaissance (ISR) Command & Control platform designed to unify drone fleet management, real-time video streaming, AI-powered threat detection, and geospatial intelligence into a single, cohesive web application.

Built on a modern cloud-native architecture with Next.js, TypeScript, and real-time WebSocket/MQTT communication, OMNIWATCH OS provides surveillance teams with an unprecedented operational advantage: the ability to monitor multiple drone feeds simultaneously, receive AI-verified threat alerts in real time, execute precision flight commands, and generate actionable intelligence reports — all from a single dashboard.

This document presents the current capabilities of the platform, outlines the strategic product roadmap including 2D/3D mosaic generation, time-series geospatial analysis, field communications, and automated report generation, and articulates the value proposition for stakeholders evaluating the platform for deployment.

2. Product Overview

OMNIWATCH OS serves as the central nervous system for drone-based surveillance operations. It bridges the gap between hardware (DJI drone fleets, docking stations, remote controllers) and actionable intelligence (threat detection, incident reporting, mission analytics).

Core Value Pillars

- **Unified Command Center:** Manage entire drone fleets, live feeds, and mission operations from a single interface.
- **AI-Augmented Intelligence:** Real-time YOLO object detection with Gemini LLM threat verification reduces false positives and accelerates response times.
- **Geospatial Awareness:** Interactive maps with live drone tracking, geofencing, flight area management, and soon — 2D/3D mosaic overlays.
- **Secure & Scalable:** JWT-based authentication, role-based access, and cloud-native architecture designed for enterprise deployment.
- **Field-Ready:** Upcoming mobile companion with real-time alerts, routing, and voice/chat communications for ground teams.

3. Architecture & Technology

OMNIWATCH OS employs a dual-backend architecture with three independent real-time communication channels, ensuring optimal performance for each specialized workload.

Technology Stack

Layer	Technology	Purpose
Frontend	Next.js 15 (App Router)	SSR, API routes, edge middleware
Language	TypeScript 5	Type safety, developer productivity
UI	React 18 + Tailwind CSS 3	Component-based responsive UI
State	TanStack React Query + Context	Server state caching, client state
Forms	React Hook Form + Zod	Form management and validation
Mapping	MapLibre GL JS	Geospatial visualization
Real-Time	WebSocket + MQTT + Socket.IO	Telemetry, flight control, AI events
Video	WebRTC (WHEP Protocol)	Low-latency drone livestreams
HTTP	Axios	API communication with auto-retry
Storage	MinIO / S3	Media and log file storage

Backend Services

Service	Technology	Responsibility
OmniWatch API	Django / Python	Auth, projects, teams, org, alerts
DJI Cloud API	Go / Cloud-Native	Devices, flight control, waylines, HMS
AI Detection	Python (YOLO + Gemini)	Object detection, threat verification
MQTT Broker	Mosquitto	Dock telemetry, DRC flight control
WHEP Server	SRS	WebRTC streaming gateway

Real-Time Communication Architecture

- Channel 1:** Native WebSocket — DJI Cloud telemetry (device status, OSD, HMS, task progress, flight area sync)
- Channel 2:** MQTT over WebSocket — Dock OSD telemetry and Drone Real-Time Control (DRC) for emergency stop and velocity commands at 50ms intervals
- Channel 3:** Socket.IO — AI Detection backend streaming YOLO_DETECTION and TRACK_CONFIRMED events

4. Current Features

The following sections detail every feature currently implemented and operational in OMNIWATCH OS.

4.1 Authentication & User Management

- **Login:** Custom email + PIN authentication (not traditional passwords)
- **Security:** JWT-based sessions with in-memory token storage (never localStorage)
- **Session:** Proactive token refresh 60 seconds before expiry; HttpOnly refresh cookies
- **Protection:** Edge middleware route protection with automatic redirect
- **Resilience:** Auto-retry on 401 with silent token refresh and request replay
- **User Management:** Organization-level staff CRUD (add, update, deactivate)
- **Invitations:** Team invitation workflow with DJI workspace provisioning
- **Roles:** Role-based access: Admin, Commander, Operator, Analyst

4.2 Dashboard & Fleet Overview

- **Fleet Stats:** Real-time fleet statistics: total drones, online drones, active missions, total media
- **Missions:** Recent missions table with status badges
- **Fleet Status:** Device online/offline status panel
- **Project Info:** Project info panel: device count, flight areas, missions, failed missions
- **Navigation:** Collapsible sidebar with grouped navigation
- **Project Context:** Active project context bar with project switching
- **Theming:** 6 premium dark themes (Dark, Midnight Phantom, Cyber Trench, Deep Matrix, Solar Eclipse, Absolute Zero)

4.3 Live Video Streaming

- **Multi-Feed:** Multi-stream grid view supporting multiple simultaneous drone feeds
- **Streaming:** WebRTC player using WHEP protocol for ultra-low latency
- **Camera Control:** Camera lens switching (zoom, wide, infrared)
- **Quality:** Stream quality selection and resolution switching
- **Discovery:** Device sidebar for browsing available streams
- **Resource Mgmt:** Automatic stream cleanup on component unmount

4.4 Geospatial Intelligence (GIS)

- **Map Engine:** MapLibre GL JS interactive map with multiple basemaps

- **Live Tracking:** Real-time drone position tracking with live telemetry overlays
- **Drone Info:** Drone info popups (battery, altitude, speed, heading)
- **Map Elements:** GeoJSON map elements: Points, LineStrings, Polygons
- **Geofencing:** Flight areas (geofences) — circle and polygon types
- **Area Sync:** Flight area sync to devices with status monitoring
- **Waylines:** Wayline visualization on map

4.5 AI-Powered Threat Detection

OMNIWATCH OS features a dual-panel AI detection system combining real-time YOLO object detection with Gemini LLM threat verification.

- **Detection Engine:** Real-time YOLO object detection with bounding box overlays
- **Panels:** Dual-panel: Orange (live YOLO alerts) and Red (Gemini-verified threats)
- **Detection Cards:** Detection cards with bounding boxes, confidence scores, track IDs
- **GPS Intelligence:** Dual GPS: drone position + estimated object position
- **LLM Verification:** Gemini LLM threat verification with reasoning explanations
- **Workflow:** Detection status workflow: pending → approved/dismissed
- **Bulk Ops:** Bulk status updates for efficient operator validation
- **Alerts:** Sound alerts for high-confidence threats
- **Mapping:** Detection map — plot detections on geospatial map

4.6 Flight Control & Mission Execution

The Mission Control interface provides full-spectrum drone command capabilities.

- **Flight Commands:** Takeoff to point, fly to point, return to home
- **Safety:** Emergency stop via DRC MQTT channel
- **Precision Control:** Velocity control at 50ms intervals
- **Camera Systems:** Gimbal control (pitch, yaw, zoom, camera mode)
- **Dock Operations:** Cover open/close, drone open/close
- **Diagnostics:** Debug mode activation for remote troubleshooting
- **DRC Protocol:** MQTT-based bidirectional DRC with 1s heartbeat

4.7 Device & Firmware Management

- **Inventory:** Device listing (drones, docks, RC controllers)
- **Binding:** Device binding/unbinding to workspace
- **Topology:** Device topology tree (dock → drone → RC)
- **Status:** Online/offline status monitoring (30s refresh)
- **Firmware:** OTA firmware updates with progress tracking
- **AI Config:** Per-device AI detection configuration

4.8 Mission Planning & Wayline Management

- **Wayline Files:** KMZ wayline file upload and management
- **Parsing:** KMZ parsing: waypoints, altitude, speed, survey polygon, duration
- **Task Creation:** Create flight tasks: immediate, timed, conditional
- **Task Status:** Task status tracking: pending, in-progress, complete, failed, paused, cancelled
- **Task Control:** Pause/resume task execution
- **Progress:** Task progress monitoring via WebSocket

4.9 Media Management

- **Media Browser:** Paginated media file listing with details
- **Download/Preview:** Media file download and preview
- **Upload:** Fast upload check with MD5 deduplication
- **Direct Upload:** Direct-to-storage upload via STS credentials

4.10 Project & Organization Management

- **Projects:** Project CRUD (create, read, update, delete)
- **Device Assignment:** Device assignment to projects
- **Project Switching:** Active project selection with session persistence
- **Organization:** Organization profile management
- **Team:** Member listing and staff management

4.11 Health Monitoring System (HMS)

- **Messages:** Workspace-wide HMS message listing
- **Unread Counts:** Per-device unread message counts
- **Filtering:** Message level filtering (info, warning, critical)
- **Real-Time:** Real-time HMS event streaming via WebSocket

4.12 Reporting & Analytics (Current)

- **Fleet Stats:** Dashboard-level fleet statistics and mission counts
- **Detection Stats:** Detection statistics bar
- **Mission Analytics:** Mission status breakdown

5. Future Roadmap

The following sections outline planned features that will transform OMNIWATCH OS from a command & control platform into a complete intelligence ecosystem.

5.1 2D/3D Mosaic Generation from Drone Imagery

A cornerstone feature is the automated generation of 2D orthomosaic maps and 3D textured meshes from drone-captured imagery, overlaid directly on the geospatial map for every mission.

Capabilities

- **2D Mosaic:** Automated 2D orthomosaic stitching from overlapping drone images
- **3D Mosaic:** 3D textured mesh generation for terrain and structure visualization
- **Mission Overlay:** Per-mission mosaic overlay on geospatial map
- **Versioning:** Time-stamped mosaic layers for before/after comparison
- **Performance:** Tile-based progressive loading for large datasets
- **Export:** Export to GeoTIFF, OBJ, glTF for external GIS tools

Use Cases

- Construction site monitoring — track progress over weeks/months
- Agricultural land assessment — crop health and irrigation analysis
- Security perimeter inspection — before/after incident comparison
- Disaster response — damage assessment and recovery tracking
- Urban planning — 3D visualization of development areas

5.2 Time-Series Geospatial Analysis

Building on mosaic generation, OMNIWATCH OS will provide time-series analysis tools to track how surveyed areas change over time.

- **Timeline Navigation:** Timeline slider to navigate mosaic snapshots across dates
- **Change Detection:** Change detection algorithms highlighting areas of modification
- **Alerts:** Automated alerts when significant changes detected in monitored zones
- **NDVI Analysis:** Vegetation index (NDVI) time-series for agriculture
- **Volumetrics:** Volumetric measurements for earthwork and stockpile monitoring
- **Activity Heatmaps:** Heatmap overlays showing activity patterns over time

5.3 Real-Time Field Alerts with Routing & Network Analysis

Field operators will receive real-time incident alerts with intelligent routing and network analysis to optimize response times and resource deployment.

- **Push Alerts:** Push notifications for AI-verified threats to mobile devices
- **Incident Mapping:** Incident location on field operator map with GPS coordinates
- **Route Optimization:** Optimal routing from operator position to incident location
- **Network Analysis:** Network analysis for multi-operator deployment
- **Escalation:** Escalation chains based on severity and operator proximity
- **Fleet Tracking:** Real-time operator position tracking for fleet coordination
- **Geofence Alerts:** Geofence-based automatic alerts in hazard zones

5.4 Field Communications (Chat & Voice)

A comprehensive communication suite enabling all field users to coordinate in real time through text and voice channels.

- **Text Chat:** Real-time text chat between field users and command center
- **Group Channels:** Group channels organized by mission, team, or incident
- **Voice Calls:** Voice call capability (WebRTC peer-to-peer or SFU)
- **Push-to-Talk:** Push-to-talk mode for quick field communications
- **Media Sharing:** Media sharing in chat (images, video clips, location pins)
- **Encryption:** End-to-end encryption for sensitive communications
- **Offline Support:** Offline message queue with sync on reconnection
- **Incident Chat:** Auto-create chat rooms per incident

5.5 On-the-Fly Report Generation

OMNIWATCH OS will generate comprehensive, professional reports per project on demand, combining mission data, detection analytics, imagery, and geospatial context.

- **Templates:** Template-based report generation (mission summary, incident, compliance)
- **Auto-Populate:** Auto-populated: mission timeline, detections, media, map overlays
- **Imagery:** Embedded 2D/3D mosaic imagery with annotations
- **Detection Analytics:** Detection statistics with confidence breakdowns
- **Flight Data:** Flight path visualization with telemetry overlays
- **Export Formats:** Export to PDF, DOCX, and printable formats
- **Scheduling:** Scheduled report generation (daily, weekly, per-mission)
- **Branding:** Branded reports with organization logo
- **Sharing:** Shareable report links with expiration and access control

5.6 Advanced Device & Firmware Management

- **Version Inventory:** Firmware version inventory across entire fleet

- **Batch Updates:** Batch OTA firmware updates with rollback capability
- **Compatibility:** Firmware compatibility matrix (device model vs. version)
- **Scheduling:** Scheduled update windows with operator notification
- **Predictive Maint:** Device health diagnostics with predictive maintenance alerts
- **Battery Mgmt:** Battery cycle tracking and replacement recommendations
- **Calibration:** Sensor calibration status and recalibration triggers
- **Usage Analytics:** Device usage analytics (flight hours, mission count)

5.7 AI Model Expansion

- **Custom Models:** Custom YOLO model training on customer-specific object classes
- **Thermal AI:** Thermal/infrared detection for night operations
- **Anomaly Detection:** Anomaly detection for perimeter surveillance
- **Collaborative AI:** Multi-drone collaborative detection
- **Edge AI:** Edge AI processing on DJI Dock

5.8 Multi-Tenant & Enterprise Features

- **Multi-Tenant:** Multi-tenant architecture with workspace isolation
- **SSO:** SSO integration (SAML, OAuth2, LDAP)
- **Audit Logs:** Audit logging for compliance
- **Custom Roles:** Custom role definitions with granular permissions
- **API Access:** API access for third-party integrations
- **White Label:** White-label deployment options

6. Integration Ecosystem

OMNIWATCH OS is designed as a hub connecting multiple specialized services into a unified operational picture.

Integration	Protocol	Purpose
DJI Cloud API	REST + WebSocket	Devices, flight control, waylines, HMS
DJI MQTT Broker	MQTT over WS	Dock telemetry, DRC flight control
AI Detection	Socket.IO	YOLO detection event streaming
Gemini LLM	REST API	Threat verification and reasoning
MapLibre GL JS	Client-side	Geospatial visualization
MinIO / S3	REST API	Media and log file storage
SRS WHEP Server	WebRTC	Video streaming gateway
OmnWatch API	REST + JWT	Auth, projects, teams, alerts

7. Security Architecture

Authentication & Session Security

- **Token Storage:** JWT tokens in-memory only — never localStorage
- **Cookie Security:** HttpOnly refresh cookies for session persistence
- **Token Refresh:** Proactive token refresh 60s before expiry
- **Auto-Recovery:** Silent 401 retry with auto token refresh
- **Route Guard:** Edge middleware route protection

Network Security

- **Headers:** X-Frame-Options: DENY — prevents clickjacking
- X-Content-Type-Options: nosniff
- Referrer-Policy: strict-origin-when-cross-origin
- Permissions-Policy: microphone=()
- **CORS:** Next.js API routes proxy to eliminate CORS exposure
- **Encryption:** TLS encryption for all communications

8. Target Users & Personas

Role	Primary Tasks	Key Features Used
Surveillance Operator	Monitor feeds, validate detections	Live Feed, AI Detection, Alerts
Field Commander	Oversee operations, review incidents	Dashboard, Control, Reports
Drone Pilot	Execute missions, control drones	Control, Tasks, Waylines, DRC
Report Analyst	Compile detection data	AI Detection, Media, Reports
System Admin	Manage users, devices, settings	Projects, Members, Assets
Field Operator	Respond to incidents	Mobile App, Alerts, Routing, Comms

9. Value Proposition

For Security & Surveillance Operations

- Reduce threat response time from minutes to seconds with AI-verified real-time alerts
- Eliminate blind spots with multi-drone simultaneous monitoring
- Provide irrefutable evidence with timestamped detection crops and video
- Optimize field deployment with intelligent routing and network analysis

For Infrastructure & Construction

- Automated progress tracking with 2D/3D mosaic comparison
- Volumetric measurements for earthwork and stockpile management
- Compliance documentation generated automatically

For Agricultural Applications

- Time-series NDVI analysis for crop health monitoring
- Irrigation assessment through multi-spectral imaging
- Automated field survey reports with anomaly highlighting

For Enterprise & Government

- Multi-tenant architecture with SSO and audit logging
- Scalable cloud-native deployment supporting hundreds of concurrent drones
- Compliance-ready reporting for regulatory requirements

- Open API for integration with existing C2 systems

OMNIWATCH OS transforms raw drone data into actionable intelligence. By unifying fleet management, AI detection, geospatial analysis, and field communications into a single platform, it eliminates the need for multiple disparate tools and provides a single source of truth for all ISR operations.

10. Conclusion

OMNIWATCH OS represents a paradigm shift in ISR command and control. The current platform already delivers significant operational value through its unified dashboard, real-time video streaming, AI-powered threat detection, and comprehensive flight control capabilities.

The planned roadmap — including 2D/3D mosaic generation, time-series geospatial analysis, field communications, on-the-fly report generation, and advanced firmware management — will establish OMNIWATCH OS as the definitive platform for drone-based intelligence operations.

With its modern architecture, secure design, and extensible integration ecosystem, OMNIWATCH OS is positioned to serve the evolving needs of security, infrastructure, agriculture, and government sectors worldwide.

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